

Olikbochon: Two-Branch, Evidence-Grounded, and Dictionary-Augmented Detection of Bengali LLM Hallucinations

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Abstract

We describe our system for the *Olikbochon* Bengali LLM Hallucination Detection Challenge, which asks whether a Bengali response is faithful or hallucinated, with the heaviest weight on Bangladesh-specific (C1) facts. Our central observation is that the task splits cleanly into two regimes with opposite failure modes: *context-grounded* rows are nearly solved by a substring-overlap rule plus a single LLM judge ($F1 \approx 0.90$), whereas *no-context* closed-book rows collapse to a constant-guess baseline ($F1 = 0.35$) and carry essentially all of the remaining opportunity. We attack the no-context branch with a stacked suite of complementary signals—LLM judges at two scales, a cross-lingual consistency check, and, most importantly, two *knowledge-grounded* verifiers: a 32B judge that reads answers against reranked Bengali-Wikipedia evidence with an UNSURE-safe three-way verdict, and a dictionary verifier that grounds idiom/word-meaning questions in *bn.wiktionary*—the corpus Wikipedia structurally cannot supply. An error analysis shows the field’s dominant failure is one-sided over-skepticism: true-but-obscure C1 facts are systematically flagged as hallucinated. Grounding directly targets this bias. Signals are stacked by per-branch nested cross-validation, and our honest out-of-fold estimates transfer to the public leaderboard within ~ 0.03 . The full system reaches a public leaderboard binary-F1 of **0.815**; a portable seven-signal subset reproduces it within noise fully offline inside the Phase-2 compute budget.

1 Introduction

Large language models are strikingly fluent in Bengali, a language of over 280 million speakers, and just as frequently and confidently wrong. A model can name the author of *Bidrohi* correctly in English yet misattribute it in Bengali; it can narrate the Liberation War in fluent prose with the wrong dates. These errors do not sound wrong—they sound like Bengali—which is what makes them dangerous [Ji et al., 2023, Guerreiro et al., 2023]. The *Olikbochon* challenge (“*the mirage of language*”) frames this as binary detection: given a prompt, a candidate response, and an optional source context, predict whether the response is faithful (label = 1) or hallucinated (label = 0). The leaderboard scores binary F1 on the hallucinated class, weighted toward the **C1** cultural-distance band where the Bangladeshi answer diverges from a globally dominant default—the band where the headline phenomenon concentrates.

Our approach rests on a single structural observation and two knowledge-grounding contributions:

- **Two-branch decomposition.** Rows with a context are a *grounding* problem and are already

near-solved by cheap signals; rows without a context are a *closed-book factuality* problem and hold the entire scoring opportunity (§3.1). Treating them separately, with a per-branch stacker, is worth more than any single model.

- **Evidence-grounded judging.** A 32B judge that reads each answer against *reranked* Bengali-Wikipedia passages, emitting an UNSURE-safe three-way verdict, is our strongest closed-book signal and rescues obscure C1 facts that every parametric judge misses (§3.3).
- **Dictionary grounding (the “right book”).** Roughly one in seven no-context test rows are idiom/word-meaning questions that Wikipedia *provably* cannot answer—dense retrieval returns grammar and physics articles. Grounding them in *bn.wiktionary* instead lifts idiom precision to 0.92 and is applied as a targeted override (§3.4).

The result is a calibrated, per-branch stacked ensemble whose honest out-of-fold (OOF) score tracks the leaderboard tightly and whose portable subset runs offline within the code-competition limits.

2 Task and Data

The organizers release a **sample split of 299 labeled rows** for pipeline validation—explicitly not a conventional training set—and a **test set of 2,516 unlabeled rows**. Each row has `prompt_bn`, `response_bn`, and a context that is `[NULL]` for closed-book items. Of the 2,516 test rows, **1,155 (46%) are no-context**. Cultural-distance bands structure the difficulty: **C0** globally stable facts, **C1** culturally situated (Bangladesh-specific), and **C2** contested or time-sensitive facts. Because there is no training signal beyond 299 examples, the design goal is *generalization*, not fitting: every modeling decision is judged on a 5-seed $\times$ 5-fold OOF estimate, and only decisive gains are promoted to a leaderboard check.

Our internal compass is **macro-F1** on the 299 labeled rows; the official leaderboard metric is binary F1 on the hallucinated class. We report both and, crucially, verify that our OOF movements survive the transfer to the hidden split (§4).

3 Method

3.1 Two-branch decomposition

Splitting the 299 samples by presence of context exposes two nearly disjoint problems. A trivial baseline—substring-overlap grounding for context rows, constant guessing for the rest—already scores macro-F1 0.894 on the context branch but only 0.345 on the no-context branch (Table 1, row 1). Context rows are extractive: a short answer either appears in, or contradicts, the passage, and substring overlap plus one LLM judge captures almost all of the signal. No-context rows require the model to *know* the answer. We therefore route every row to a branch and fit a *separate* stacker per branch, so that a signal earning high weight on closed-book factuality is never diluted by the context rows it cannot help.

3.2 Signal suite

Each signal outputs a per-row $P(\text{faithful}) \in [0, 1]$.

Cheap grounding signals. *Substring* overlap between response and context (the dominant context-branch signal); and an *NLI* check with mDeBERTa-v3-XNLI [He et al., 2021, Laurer et al., 2024] using the context as premise. NLI *under*-performs sub-

string on the context branch (0.635 vs. 0.894)—a short extractive answer is a poor NLI hypothesis—so it is retained only as a decorrelated minor feature.

LLM judges. We use Qwen2.5-Instruct [Qwen Team, 2024] at two scales as GPTQ-Int4 models via vLLM [Kwon et al., 2023, Frantar et al., 2023]. Judges differ in prompt, quantization, and decoding, which decorrelates their errors: a 14B self-consistency judge; a 14B answer-then-compare self-verifier; a 32B single-pass judge; and, to counter over-skepticism (§5), a 32B *self-verify* judge with a three-way YES/NO/UNSURE head and few-shot exemplars that include true-but-obscure local facts. Because Bengali is ≈ 1 token per character in Qwen’s vocabulary, all Bengali-heavy prompts run at `max_model_len=8192`.

Cross-lingual consistency. We ask the model in English and check agreement with the Bengali answer. Notably, on $\sim 44\%$ of no-context questions the model abstains *in English too*: these deep-C1 facts are beyond parametric recall in any language, which is precisely why external knowledge is needed.

3.3 Evidence-grounded 32B judging

Our strongest closed-book signal reads each answer against retrieved evidence rather than the model’s parameters. We build a Bengali-Wikipedia index from the current bnwiki dump (438,788 articles $\rightarrow$ 335,628 passages), retrieve with dense **bge-m3** [Chen et al., 2024], rerank with **bge-reranker-v2-m3**, and pass the top passages to the 32B judge. The judge emits a three-way verdict; critically, **UNSURE maps to 0.5** rather than a wrong NO, so missing evidence abstains instead of manufacturing a hallucination call. This single signal reaches standalone macro-F1 0.753 (previous best 0.688), is decisive on every row, and rescues 7 of the 23 faithful rows that *every* other signal had missed. It earns the largest no-context stacker weight in the ensemble.

3.4 Dictionary grounding: the right book

Error analysis (§5) revealed a class of rows no amount of Wikipedia retrieval can solve: idiom and word-meaning questions. Asked the figurative sense of *ujaner koi* (‘an easily obtained thing’) or *dhandha* (‘a dishonest scheme’), dense retrieval over Wikipedia

returns grammar, physics, and fruit articles—an encyclopedia simply does not contain idiom glosses. And this is not a corner case: **162 of the 1,155 no-context test rows** (14%) are word/idiom/meaning questions (75 idiom “bhabartha” rows, 80 word-meaning, plus synonyms/antonyms/proverbs).

The fix is the correct corpus. We parse `bn.wiktionary` with a custom parser (`wikiextractor` targets article prose, not lexical entries) into **73,444 entries**—15,182 idioms (including 7,723 mined from the *bagdhara* appendix table) and 58,262 words—keyed by headword and definition. Dense `bge-m3` retrieval recovers variant spellings that exact lookup misses, and the 32B judge grounds the given meaning against the retrieved definition with the same UNSURE-safe protocol.

Accuracy splits sharply by question type. On **idioms the signal is 12/13 = 0.92**: it passes true idiom glosses and rejects false ones, and grounds all 75 idiom test rows decisively. On raw word-meaning it is only 0.50, wrecked by dataset label noise (definitions that match the dictionary yet are labeled hallucinated) and polysemy. We therefore deploy it not as a stacker feature (too sparse to weight) but as a **targeted idiom-only override**: where the dictionary verdict is confident on a *bhabartha* row, it overrides the ensemble. On the test set it flips 26 of 75 idiom predictions (net +18 faithful).

3.5 Per-branch stacking

Signals are combined by **per-branch logistic regression** refit at run time on the 299 labeled rows, with weights and decision thresholds estimated by 5-seed $\times$ 5-fold nested cross-validation (the threshold is the CV median). The stacker allocates the top no-context weight to evidence grounding, with the 32B self-verify and cross-lingual signals contributing complementary weight; the context branch is carried by substring and the judges. Adding *more* parametric judges past this point does not help—agreement among judges is 0.74–0.86 and the ensemble saturates—so gains come only from new *knowledge* (retrieval, dictionary), not new model scale.

4 Results

Table 1 traces the honest OOF macro-F1 and the corresponding public-LB binary-F1 as signals accumulate. The no-context branch moves from a constant-

System	OOF	no-ctx	LB
Substring + constant (baseline)	0.682	0.345	0.666
+ 14B judge	0.765	0.649	—
+ cross-lingual	0.779	0.670	—
+ v1 retrieval (5 sig.)	0.793	0.675	0.752
+ 2nd judge & self-verify (7 sig.)	0.795	0.708	0.765
+ 32B judge (8 sig.)	0.803	0.717	0.759
+ 32B self-verify + ret32	0.833	0.778	0.803
+ wiktionary override	0.846	—	0.815

Table 1: Honest OOF macro-F1 (5-seed $\times$ 5-fold) and public-LB binary-F1 as signals accumulate. Dashes: not separately submitted. Evidence and dictionary grounding drive the final gains.

guess 0.345 to ~ 0.78 ; the context branch stays near 0.90 throughout. Two facts matter more than any single number. First, the **OOF $\rightarrow$ LB transfer is tight** (≈ -0.03 at every checkpoint): the 299-row harness is a trustworthy compass despite its size. Second, the last two gains come entirely from grounding—evidence (`ret32`) and the dictionary override—not from stacking additional parametric judges.

The evidence-grounded 32B signal (`ret32`) lifts OOF from 0.803 to 0.833 ± 0.008 and the public LB from 0.759 to 0.803. The dictionary override adds a further +0.013 OOF (0.846 ± 0.008) and +0.012 LB ($\rightarrow 0.815$), rescuing ~ 4 of the 16 hardest rows. Pruning two saturated signals (NLI, cross-lingual) is a further variance-reducing +0.005 OOF hedge.

5 Analysis

The failure mode is one-sided. On the 167 no-context sample rows, the 23 rows that *every* signal gets wrong (or abstains on) are **all faithful**. Not a single hallucinated row evades all signals. Our judges are systematically *over-skeptical*: true but obscure C1 answers get called hallucinated. This one-sided bias—not random error—is the gap to the top of the leaderboard, and it is exactly what evidence and dictionary grounding target.

Autopsy of the residual. After `ret32`, 16 hard faithful rows remain. Their taxonomy is instructive: 5 are idiom/proverb rows that defeat every parametric judge (all ≈ 0) and are solvable *only* by the `wiktionary` corpus; 5 are literature/geography/history facts needing deeper Wikipedia retrieval; 4 are

Model	Role	Disk
Qwen2.5-14B-Instruct GPTQ	judge, self-verify	~9 GB
Qwen2.5-32B-Instruct GPTQ	judge, grounding	~19 GB
BAAI/bge-m3	dense retrieval	~2.3 GB
BAAI/bge-reranker-v2-m3	reranking	~2.3 GB
mDeBERTa-v3-base-xnli	NLI	~1.1 GB
bnwiki + bn.wiktionary	evidence index	~2 GB

Table 2: Open-weight models. Total ~37 GB, under the 50 GB limit; runs offline in under 9 h on 2×T4.

parametric-recall rows the anti-skepticism judges already read correctly; 1 is an underspecified negation MCQ with its options stripped; and 1 is **genuine label noise**—*competent* (‘unqualified’) is the exact opposite of the truth, so the judges are *correctly* flagging it. Chasing that last row would mean teaching the model a falsehood, so we exclude it. A useful byproduct: the dictionary *surfaced* several mislabeled word-meaning rows, a real ceiling on that slice we reported to the organizers rather than fit.

Saturation. Once three grounding-and-judging angles are in the stack, adding further judges hurts within noise. On a 299-row validation set, OOF deltas under ~0.02 do not reliably transfer; we treat them as noise and do not chase them in either direction.

6 Reproducibility and Phase-2 Package

The full system uses ten signals, three of which were computed on a personal workstation. For the offline code-competition run we ship a **portable seven-signal subset** that reproduces the result within noise (OOF 0.831 vs. 0.833; the difference is far below a 299-row resolution) as a single Kaggle notebook. It recomputes every signal from the raw test CSV for both the 299 labeled rows (to fit the stacker) and the evaluation rows, then writes `submission.csv`. Table 2 lists the weights; all are open, total ~37 GB (<50 GB), and GPTQ (not AWQ) quantization is deliberate so the same weights run on both P100 (sm60) and T4 (sm75). A verification run on the Phase-1 test set completed offline in **4.96 h** on 2×T4—well inside the 9 h budget—and reproduced our submitted predictions.

7 Conclusion

The Bengali hallucination-detection task is not one problem but two, and its dominant failure is a one-sided skepticism toward true-but-obscure culturally-situated facts. Progress came not from larger judges but from giving the judge the *right evidence*: reranked Wikipedia passages for factual recall, and—where an encyclopedia is structurally silent—a Wiktionary idiom dictionary. Both are cheap, offline, and directly aimed at the C1 band the task rewards. The system reaches a public-LB binary-F1 of 0.815 with an honest validation harness that transfers to within 0.03 of the leaderboard, and it reproduces fully offline within the Phase-2 compute budget.

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